

Yumeng He

+1-801-347-7770 | u1528477@umail.utah.edu |  ymh1003

Salt Lake City, UT

EDUCATION

- **University of Utah** August 2024 - Present
PhD in Computer Science Salt Lake City, UT
 - GPA: 4.00/4.00, advised by Pavel Panchekha
 - Coursework: Programming Languages, Operating Systems, Computer Architecture, Software Verification
 - Research Interests: Program analysis, numerical computing, and formal verification.
- **University of Rochester** August 2020 - May 2024
Bachelor of Science, Double Major in Computer Science and Mathematics, Minor in Business Rochester, NY
 - Grade: 3.98/4.00, graduated summa cum laude, highest distinction

PUBLICATIONS

- [1] **He, Y., & Panchekha, P.** (2026). Accurate Residues for Floating-Point Debugging. *OOPSLA 2026* (To Appear).
- [2] **He, Y., Nandi, C., & Pai, S.** (2025). Formalizing Linear-Motion G-code for Invariant Checking and Differential Testing of Fabrication Tools. *OOPSLA 2025*. [[DOI](#)] [[Talk](#)] [[Code](#)]

PROFESSIONAL EXPERIENCE

- **Annapurna Labs, AWS** May 2026 - Aug 2026
ML Compiler Engineer Intern Cupertino, CA
 - Developed compiler support and optimizations for dynamic tensor shapes on AWS Trainium and Inferentia chips.
 - Designed and implemented loop-invariant code motion with SSA-based safety and register pressure analyses.
 - Built IR and register allocation support for hardware shape registers.
 - Extended common subexpression elimination to eliminate redundant shape register computations.

PROJECTS

- **Efficient Numeric Debugging with Improved Accuracy [1]** August 2024 - Present
Skills: C/C++, dynamic analysis, runtime instrumentation, floating-point arithmetic, numerical analysis
 - Improved diagnostic accuracy in an LLVM-based floating-point debugger without high-precision arithmetic.
 - Extended a WebAssembly interpreter for shadow execution to track per-variable floating-point error.
 - Evaluated on 44 real-world workloads, showing higher accuracy on cases where baseline systems yield false results.
- **Differential Testing on G-code for 3D Printing Bug Detection [2]** May 2023 - October 2025
Skills: Python, static analysis, formal semantics, computational geometry
 - Developed a static analysis tool to detect software bugs in 3D printing pipelines using differential testing on G-code.
 - Defined formal G-code semantics for scalable geometric comparison.
 - Detected rotation-invariant violations across 58 real-world models, revealing bugs in production fabrication tools.
- **Lease Cache Design** May 2023 - August 2023
Skills: Rust, cache/memory systems, memory locality, probability theory
 - Modeled performance degradation resulting from optimal virtual lease assignments in physical cache systems.
 - Developed mathematical proofs for key properties of the model.

HONORS AND AWARDS

- **Undergraduate Research Award** May 2024
Department of Computer Science, University of Rochester
 - One of four recipients at the graduation ceremony.
- **Phi Beta Kappa** Elected March 2023
University of Rochester Chapter

MISCELLANEOUS

- **TA for Computer Organization, CS Dept., University of Rochester** Spring 2023
- **TA for Operations Research, Math Dept., University of Rochester** Fall 2022
- **Product Management Intern, Trucker Path, Renren Inc.** May 2021 - July 2021